(PROJECT PROPSAL)

AUTOMATIC PET FEEDER AND PET MONITORING SYSTEM THROUGH Wi-Fi

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Abstract

This senior project proposes a development of an automatic pet feeder and pet monitoring system through Wi-Fi especially on cat as the testing pet. Pet feeder device is based on several IoT technologies which is Ultrasonic, Infrared, Load weight sensor, Servo motor, Water pump and ESP32-CAM. User can feed and monitor their pet automatically through mobile application.

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CHAPTER 1
INTRODUCTION

1.1 Background and Rational

Animal feed distribution systems are a typical item that are utilized on a big scale in commercial applications as well as on a smaller scale in household applications.

Animal feed distribution systems exist in a variety of shapes and sizes, with various controls over how the feed is distributed. There are many various ways to achieve the same end result, some of which are more efficient than others, whether it be a manual system, an automatic system on a timer, or a sensor-based system. When it comes to feeding systems, the usage of sensors is a significant plus because it entirely automates the process, allowing for minimal human intervention.

Many users are looking for a system that is not only capable of working on its own, but also pleasant to the eye. Many young adults in today's collegiate society are eager for new experiences, and one of these experiences is taking a pet home. Whether it's a dog, cat, hamster, or any other type of pet, these creatures require adequate care and nutrition. Not getting fed the right amounts of feed or not getting fed on time can be a source of stress for these animals. This is a tragic fact that leads to malnutrition and, ultimately, abandonment of these creatures. Furthermore, pet food is frequently packaged in an airtight bag, which pet owners may fail to close entirely after feeding their pets. As a result, mold spores and bacteria cultures can form in the feed itself.

1.2 Objective

1.2.1 To develop automatic pet feeder and pet surveillance system through Wi-Fi using Blynk application.
1.2.2 Monitor the pet with camera through mobile application.
1.2.3 Weight the food on the plate with weight sensor.
1.2.4 Provide the correct amount of food to the pets.
1.2.5 Pet can drink the water with the water fountain automatically. The sensor will detect the pet then turn on the water fountain.

1.3 Scope

1.3.1 General user
- Pet owner

1.3.2 Setting the pet time to eat at least 3 times a day according to the amount of food depending on the breed and its species
1.3.3 Automatically use programme rules and mechanical concepts
1.3.4 Design and dedicate this machine
1.3.5 Hardware
- ESP32-CAM
- Servo motor

- Weight sensor
- TCRT Infrared
- Water pump
- Ultrasonic sensor

1.3.6 Software

- Blynk mobile Application
- Use C language in Arduino program

1.3.7 Testing

- Test with 3 cats
- Test using household wifi

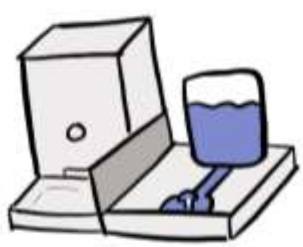
Water pump

Ultrasonic sensor

TCRT Infrared

Servo Motor

Weight Sensor



Pre-model 1.3

Figure 1.3.5 Hardware

1.4 Methodology

1.1.1 Proposal preparation and defense

- sketching a scope of project

1.1.2 Literature review and requirement gathering

- Discussing with teammate, advisor and research about project topic

1.1.3 System analysis

- Make a decision for this project

1.1.4 System design

- Design an image of product
- Design a hardware and software will be use.

1.1.5 Computer Engineering Pre-Project document preparation

- Document introduction
- Document literature review
- Document design

1.1.6 Computer Engineering Project EXAM

1.1.7 System development

- Software
- Hardware

1.1.8 System testing

- Testing software systems
- Testing Hardware systems

1.1.9 Computer Engineering Pre-Project2 exam

1.1.10 Computer Engineering Pre-Project2 exam

1.5 Plan

Table 1.1 Working plan

	Computer Engineering Pre-project					Computer Engineering Project				
Plan/Month	Jan	Feb	Mar	Apr	May	Aug	Sep	Oct	Nov	Dec
Proposal preparation and defense										
Literature review and requirement gathering										
System analysis										
System design										
Pre-project presentation										
System development										
System testing										
Project presentation										

1.6 Progression

Table 1.2 Task and Responsibility

Week	Task	Responsibility
February 1	Prepare Background and Rational	Hassanseeroyee Saeaed
	Research a methodology	Weishaanismail Noipom
	Making a plan	Safia Yahlee
February 7	Research an Objective for project	Hassanseeroyee Saeaed
	Research expected result for project	Weishaanismail Noipom
	Research place and equipment	Safia Yahlee
February 14	Scope a project	Hassanseeroyee Saeaed
	Research hardware and software for a project	Weishaanismail Noipom
	Predict a budget	Safia Yahlee

Week	Task	Responsibility
March 1	Research marketplace for animal	Hassanseeroyee Saeaed
	Research Criteria Comparison	Weishaanismail Noipom
	Research pet’s food habit	Safia Yahlee
March 8	Research automatic pet feeder market	Hassanseeroyee Saeaed
	Research related technology	Weishaanismail Noipom
	Research proper nutrition	Safia Yahlee

March 15	Research the reason for increase in automatic pet feeder	Hassanseeroyee Saeaed
	Research Arduino software and Blynk	Weishaanismail Noipom
	Research the amount of water and food needed per day	Safia Yahlee

Week	Task	Responsibility
April 4	Research electronic configuration	Hassanseeroyee Saeaed
	Research system flowchart	Weishaanismail Noipom
	Research suitable outer and inner design	Safia Yahlee
April 11	Finish designing system schema	Hassanseeroyee Saeaed
	Finish designing flowchart for the system	Weishaanismail Noipom
	Finish designing outer inner part of device	Safia Yahlee
April 18	Finish drawing system schema in program	Hassanseeroyee Saeaed
	Finish creating flowchart in the program	Weishaanismail Noipom
	Finish drawing outer and inner part of device	Safia Yahlee

1.7 Expected Result

1.7.1 Pet feeders help to ensure that your pet does not miss a meal, whether you are in the house of away on a business or any other kind of trip.

1.7.2 The device helps to ensure that your pet does not overfeed when you are away since it dispenses specific portions as programed by you. This fact is especially important because overeating may result in issues such as obesity, which would give rise to more severe medical problems.

1.7.3 Pet feeders also help to ensure that you maintain the eating schedule for the pet even when you are not physically close to the pet.

1.7.4 Pet feeder dispense food or water each time the pet comes near the device. This will help save power and maintain the device long.

1.7.5 A pet feeder helps pet owners to train their pets to eat on schedule and specific amounts of food.

1.7.6 This designs of pet feeders come with a camera on which can monitor the pets through mobile application.

1.8 Place and Equipment

1.8.1 Pet owner household

1.8.2 Hardware (Identify also the minimum specification)

1.8.3 Software (List all target software and version)

1.9 Budget

Table 1.3 Equipment and Budget

No	Materials / Equipment	Amount	Price (Baht)
1	ESP32-CAM	1	300
2	Servo Motor	1	100
3	TCRT Infrared	1	20
4	Water pump	1	200
5	Weight Sensor	1	220
6	Ultrasonic	1	40
7	Breadboard	1	40
8	Plate for Pet	2	50
9	FT232 USB	1	90
10	Acrylic Sheet	4	500
Total		14	1560

This is an estimation of the equipment that we use in this project based on the current online shopping site.

CHAPTER 2
LITERATURE REVIEW

2.1 INTRODUCTION

marketplace for animals It's another market that's popular right now since it's going against the current trend. Raising pets was not common prior to the invention of Covid. Working-age people who rarely remain at home, for example, work an average of 8 hours every day. Raising a pet, or a group of teens (in college) who enjoy going on journeys in search of new experiences, is, of course, not a good choice. However, after the occurrence of COVID-19, the pet market became a popular trend. People of all ages have been forced to stay at home owing to Covid 19. Working from home, learning online for nearly 24 hours a day, and living at home, of course, has an impact on the minds of individuals of all ages. Some people suffer from anxiety or depression. This allows folks to identify stress-relieving activities that they can undertake at home. Pets are, without a doubt, popular. During COVID-19, cats, dogs, birds, and rabbits (small animals) are the most common pets. The pet industry is rising at a rate of 63 percent.
Of course, amenities in the pet market are growing very quickly, such as automatic cat toilets, cat fountains, automatic cat feeders.

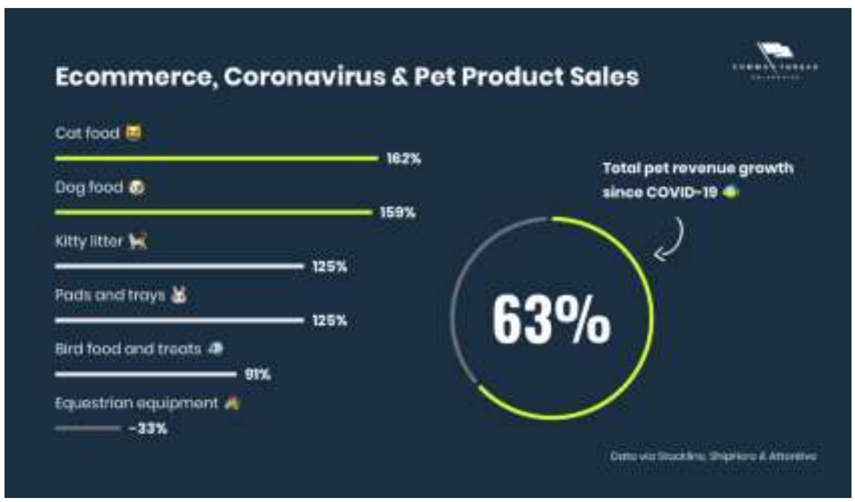


Figure 2.1 Ecommerce, Coronavirus and pet product sales

2.2 Automatic Pet Feeder Market Forecast and CAGR

You have the responsibility of providing your pet with high-quality, well-balanced food at regular intervals as a pet owner. This has fueled the demand for automatic pet feeders by increasing pet owners' concern about providing a healthy and nutritious meal for their cherished pets. According to a survey provided by Fact.MR, the automatic pet feeder industry is predicted to increase at a rapid rate between 2021 and 2031. As a result of the recent trend of maintaining pets and supplying them with timely meals even when they are not there, pet adoption rates are predicted to rise.

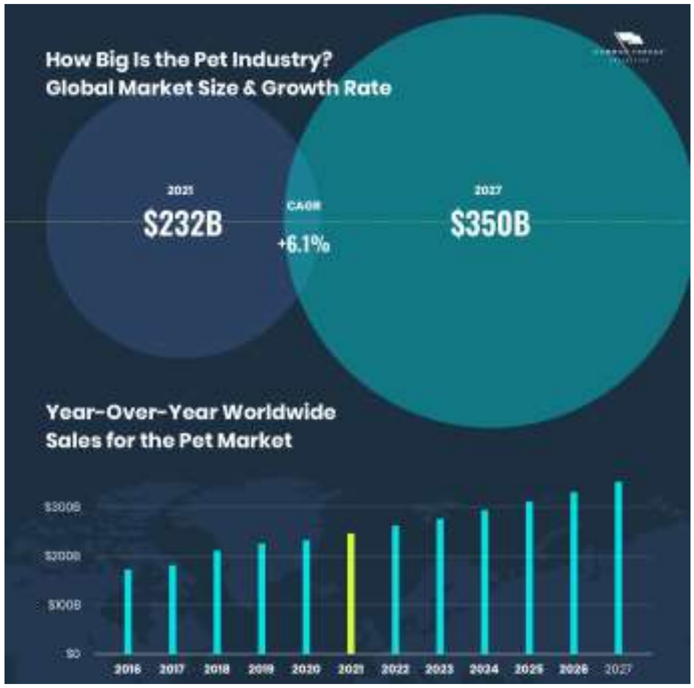


Figure 2.2 Pet Industry Market

2.3 What is the Reason for the Increase in Automatic Pet Feeder Demand?

Urban inhabitants' changing living preferences and tight work schedules, as well as an increase in product awareness and accessibility thanks to internet commerce, are driving demand. Furthermore, thanks to an increase in the conceptualization of smart homes, the adoption of automatic and smart pet feeders is strongly pushing market sales. As pet owners become more concerned about their pets' well-being, the market is gaining traction. Adoption of pets as companions for mental well-being, health, and amusement is on the rise, which is pushing increasing pet-care spending. As a result, manufacturers are continuing to profit from the introduction of smart gadgets. These products not only supply timely food to pets, but they also keep track of a variety of other factors, which they may access via a smartphone companion app. Furthermore, as pet owners become more concerned about a balanced and timely diet as a result of improved understanding about pet health, demand is likely to rise in the coming years. Furthermore, ongoing technology improvements in emerging nations, such as the use of a video camera to enable adequate pet tracking, are projected to boost the adoption of automatic and smart pet feeders in the future.

2.4 Pet's Food Habit

Pet	Pet's Food Habit Information
Hamster 	Natural food for hamsters Usually a variety of grains such as corn, pumpkin seeds, various nuts, almonds and there are also fruits and vegetables. However, eating large amounts of fresh fruits and vegetables can cause diarrhea in your hamster. It should be given in small amounts and infrequently as a dietary supplement. The best way is to choose dried vegetables and fruits. Snacks for hamsters Suitable for adult rats in snacks, they are often rich in different nutrients such as vitamins and minerals to make hair shiny, help strengthen the immune system, etc.
Rabbit 	The right food for rabbits is the food that wild rabbits eat in nature is fresh grass that consists of Crude fiber about 20-25 percent, crude protein (crude protein) about 15 percent and fat about 2-3 percent will be fresh grass. or good quality but should be available to eat at all times Green leafy vegetables, if they are to be eaten, must be washed thoroughly before giving. There are many types of ready-made food for rabbits. both mixed food It consists mainly of cereals, legumes, and various forms of pellet food, either in pellets or by heat and pressure. Sometimes alfalfa is added to add fiber and calcium. In the latter phase, pre-packaged foods are usually pellets. or pellet food that has gone through more heat and pressure
Dog 	Puppies 2 months old should be fed 4 meals a day. Puppies 3 months old should be fed 3 meals a day. Puppies 4 months – 1 year old should be fed 2 meals a day. Dogs over 1 year old should be given 1-2 meals a day. Add a few tips that always feed in one bowl and the same area. To create discipline for eating the right food with the dog.
Bird 	How do we eat all the food categories for good health? Birds need the same protein, fat, fiber, carbohydrates, vitamins and minerals to grow. Maintain good health and reproduction. Vegetables and fruits that contain fresh vitamins and nutrients. is very good for birds So what are the fruits and vegetables that have vitamins and nutrients that are good for them? -Types of melons Melon, cantaloupe, watermelon, even pumpkin is rich in fiber. which is good for the digestive system of birds as well as being high in vitamin C - Berry Give the bird berries 2-3 times a week.

	- Nuts and sprouts Beans and sprouts of various vegetables You can feed the birds as often as you like. These foods are high in protein. - Carrot Carrots contain vitamins that will help birds have sharp eyesight.

2.5 Proper Nutrition

2.5.1 Kitten age

A kitten's stomach is also small and cannot function as fully as an adult cat. This makes him eat less per meal and may not be getting enough nutrients for his body's growth. Therefore, everyone should split the kitten's meal into smaller meals. but to eat many meals during the day. You can divide the frequency of meals for cats according to their age as follows.

After the kittens are weaned and aged between 3-6 weeks, give him a kitten food formula in the amount of 1/4 - 1/3 cup per day (29 - 39 g). During the 3-4 week old kitten's milk teeth. the cat will fall out May cause him to hurt its gums until he eats less.

After the kittens are 7-12 weeks old, continue to feed them kitten formula cat food. Continue with 1/3 - 3/4 cup daily (39 - 88 g), but reduce meal frequency to 2 meals per day.

When the kitten is 24-25 weeks old, it is still recommended that he continue to feed the kitten formula in the amount of 1/2 - 1 cup per day (58 - 88 grams) until he reaches the age of 1 year. You may notice that all of his permanent teeth will grow. which the selection of dry kitten food for it during this period This will help reduce plaque build-up on your cat's teeth as well.

2.5.2 Adult age

Food should be divided into 2-3 meals per day.

Cat weight(kg.)	Amount of food(gram/day)
2-3	47-61
3-4	61-74
4-5	74-86
5-6	86-97
6-7	97-108
7-8	108-118

2.5.2.1 Calculating the nutritional needs of dogs and cats

This article will discuss the calculations required for the application of nutrition knowledge for dogs and cats, including:

(1) Resting energy requirement (RER) and daily energy requirement (DER)]

(2) Comparison of granular and wet nutrients

(3) Balancing nutrients or as needed by Pearson square method

2.5.2.2 Resting energy requirement and daily energy requirement

Resting energy needs are the energy needs of the animal while at rest in a properly heated cage. energy demand
per day is the animal's daily energy needs with various daily activities such as walking, running, playing, etc., including differences in age, sex, breed
and the weight of the various activities can be calculated from the formula **70 x body weight (kg)** $^{0.75}$

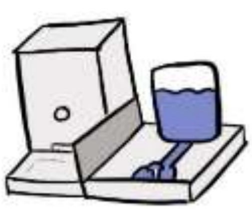
2.5.2.3 The amount of water needed per day

Water is necessary for all living creatures, including cats, to survive. Because water is the major component of cells in the body and helps to balance the body, if there is no food to live longer than dehydration. Allow each system to function properly. Cats, on the other hand, have a habit of drinking a lot of water. However, drinking too little water can lead to a variety of issues.
Drink less water Origin: with the ancestors of cats. from a desert environment This makes it possible to adapt to receive water from food in small quantities, even today the cat's body is in a different environment. But the habit of drinking less water is still following.
You must drink this water: Cats drink **roughly 60 milliliters of water per kilogram of body weight every day**. Simply expressed, a 4 kilogram cat will drink roughly 240 ml of water each day, which is about 1 glass of water.
It's problem if you don't drink enough water: If the cat drinks too little water, it may get dehydrated, which can lead to urinary system difficulties and gallstones. A minor issue will very certainly escalate into a major issue.

Cat Type

Cat Weight	Typical pet, neutered	Typical pet, intact	Typical pet, prone to gaining weight	Pet in need of weight loss
2.3 kg	157 kcal/day	183 kcal/day	131 kcal/day	105 kcal/day
3.4 kg	210 kcal/day	245 kcal/day	175 kcal/day	140 kcal/day
4.5 kg	260 kcal/day	303 kcal/day	216 kcal/day	173 kcal/day
5.7 kg	298 kcal/day	362 kcal/day	258 kcal/day	207 kcal/day
6.8 kg	354 kcal/day	413 kcal/day	295 kcal/day	236 kcal/day
7.9 kg	396 kcal/day	462 kcal/day	330 kcal/day	264 kcal/day
9.1 kg	440 kcal/day	513 kcal/day	367 kcal/day	293 kcal/day

2.6 Criteria Comparison

Diagram	Camera	Set timer	Water fountain	Easy to clean	Notified when empty	notified when cat is near	Mobile control	Manual control	Screen identification	Microphone & speaker	Adjust the control	Recommended setting	Sources
	✗	✗	✗	✓	✗	✗	✗	✓	✗	✗	✓	✗	Shopee.co.t h
	✗	✓	✗	✗	✓	✗	✗	✓	✓	✗	✓	✗	Shopee.co.t h
 Petoneer Nutri Vision	✓	✓	✗	✗	✓	✗	✓	✓	✗	✓	✗	✗	Lazada.co.th
 Our Design	✓	✓	✓	✓	✓	✓	✓	✓	✗	✗	✓	✓	

2.7 Related Technology

2.7.1 ESP32-CAM

The ESP32-CAM is a small size, low power consumption camera module based on ESP32. It comes with an OV2640 camera and provides onboard TF card slot.

The ESP32-CAM can be widely used in intelligent IoT applications such as wireless video monitoring, WiFi image upload, QR identification, and so on.

2.7.1.1 Features

- Onboard ESP32-S module, supports WiFi + Bluetooth
- OV2640 camera with flash
- Onboard TF card slot, supports up to 4G TF card for data storage
- Supports WiFi video monitoring and WiFi image upload
- Supports multi sleep modes, deep sleep current as low as 6mA
- Control interface is accessible via pinheader, easy to be integrated and embedded into user products

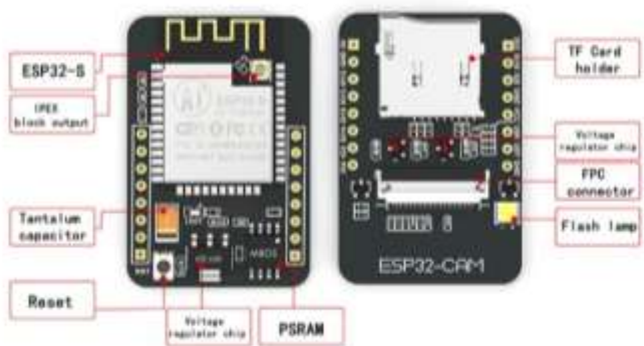


Figure 2.7.1.1 ESP32-CAM

Figure 2.7.1.2 ESP32-CAM GND

2.7.2 OV2640 camera

OV2640, an evergreen image sensor from OmniVision. OV2640 Camera is a 2 Megapixel OV2640 camera module with an f3.6mm lens. It contains a 24pin FPC interface. A wide-angle lens and 1632 x 1232 high resolution make it a perfect camera for Grove AI HAT for Edge Computing and other Sipeed serial board.

2.7.2.1 Camera Features:

- 2 Megapixel
- Array size: UXGA 1622X1200
- Power supply: 3.3V
- IO voltage level: 1.7V~3.3V DC
- Output formats:
    1. YUV(422/420)/YCnCr422
    2. RGB565/555
    3. 8-bit compressed data
- Max image transfer rate:
    1. UXGA/SXGA 15fps
    2. UXGA/SXGA 30fps
    3. SVGA 30fps
    4. CIF 60fps

Figure 2.7.2 OV2640 camera

2.7.3 Arduino Software (IDE)

The open-source Arduino Software (IDE) makes it easy to write code and upload it to the board. Sending a set of instructions to the board's microcontroller through Arduino software allows us to tell microcontroller board what to do. To do so the Arduino programming language need to be use in this program to initiate the command to the board. Software interface is shown in Figure 2.7.3

Figure 2.7.3 Example of Arduino Software

2.7.4 Blynk

Blynk is a platform that allows the user to quickly build interfaces for controlling and monitoring hardware projects from Mobile device. This application can create a project dashboard and arrange buttons, slider s, graphs, and other widgets onto the screen.

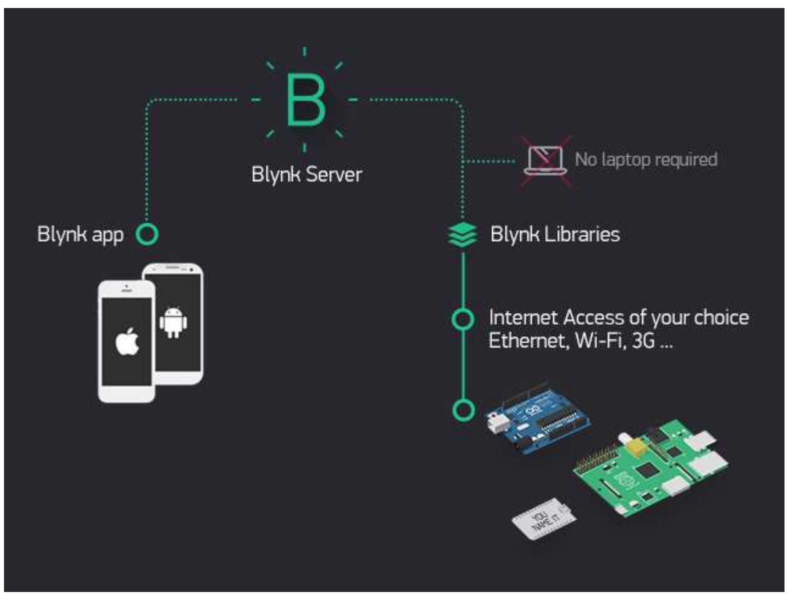


Figure 2.7.4 Blynk Operation

2.7.5 Servo Motor

A servomotor is a linear actuator or rotary actuator that allows for precise control of linear or angular position, acceleration, and velocity. It consists of a motor coupled to a sensor for position feedback. It also requires a relatively sophisticated controller, often a dedicated module designed specifically for use with servomotors.

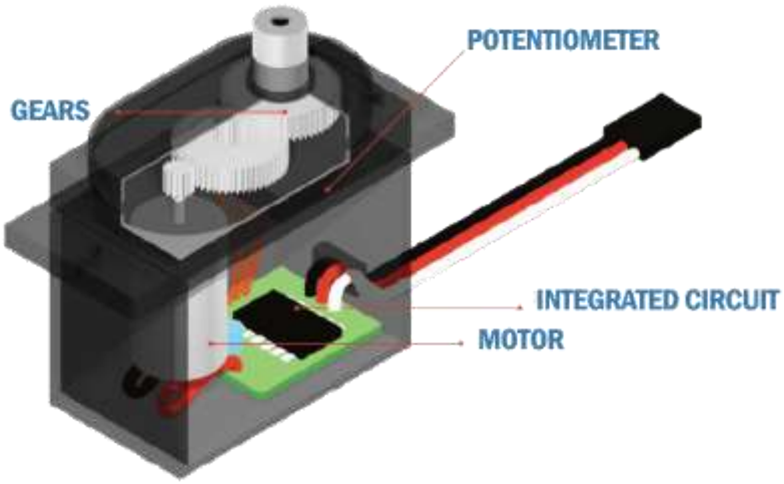


Figure 2.7.5 Servo Motor

2.7.6 Load Cell Weight Sensor

A load cell is a transducer which converts force into a measurable electrical output. Although there are many varieties of force sensors, strain gauge load cells are the most commonly used type.

Figure 2.7.6 Load Cell Weight Sensor

2.7.7 Water pump

The automated pet feeder will have feature of having a water reservoir that will pump water into a bowl that way the pet will always have access to a fresh supply of water. If the pets bowl is getting low on water, a signal will be sent to the microcontroller to turn on the water pump that will pump water from the reservoir to the bowl. Some features to take in account would be rate of displacement, size, noise level, and power rating. The pump needs to be able to pump the water at slow enough rate that gives the scale below the bowl, enough time to calculate how much water has been dispensed and signal weather to continue dispensing water or to shut off the water pump. Another attribute is that the pump needs to be able to hold back any water from leaking into the bowl when the pump is in standby mode. Any leaks can lead to an overflow of water into the bowl and potentially onto the resting surface of the feeder. The pump must also operate at a low level of noise. Like the motors that will be selected for the food dispenser and the food door cover, the pump cannot operate at a level of noise that will alarm the pet. The feeder cannot cause an adverse reaction based on loud operating motors or moving parts. Lastly the pump must also operate at or below the 24 DC volts that are available to the pet feeder system.

Figure 2.7.7 Water Pump

2.7.8 Ultrasonic sensor

After through research, an ultrasonic proximity sensor would be the best fit for the Automated Pet Feeder. The pet does not need to be in close contact with the sensor and the collar does not need any extra accessories such as metal or a magnet for detection. It could be possibly hazardous including a piece of metal or a magnet to the collar because there could be a possible that it accidently falls, and the pet can ingest it, which can be a severe liability. There are three options that were considered when it came to functionality and price are: The Ultrasonic Sensor – HC-SR04, Sparkfun RGB and Gesture Senor – APDS-9960 and the Ultrasonic Range Finder – HRLV- MaxSonar-EZ4.

The Ultrasonic Senor – HC-SR04

- have a range of 2cm to 400cm of contactless measurements.
- The device sends eight 40kHZ waves at the speed of sound and waits to detect is a pulse wave is received in the echo pin.

- The echo pin returns the time in microsecond that it took the sound wave to travel. It has four pins: the echo, trigger, ground and VCC which makes the device compact and will not take up much space on the microcontroller. It operates on 5V with a working current of .015A

Figure 2.7.8 Ultrasonic sensor

2.7.9 TCRT Infrared

The features and specifications of the TCRT5000 IR Sensor include the following.

- It is available in a leaded package
- The type of detector used is photo-transistor
- The Peak operating distance is 2.5 mm.
- Collector current ranges from 0.2 mm – 15 mm
- The typical o/p current blow test (IC) is 1 mA.
- Blocking filter for daylight
- The wavelength of the emitter is 950 nm
- Infrared sensor including the o/p of transistor
- The operating voltage is 5V
- The forward current of the diode is 60mA
- Output data is analog/digital
- The Collector current of the transistor is 100mA
- Operating temperature ranges from -25°C to +85°C
  The equivalent TCRT5000 IR sensor is RPR220 and other infrared sensors are IR LED, IR Photodiode, qtr-1rC, GP2Y0A21, TSOP, etc.

Figure 2.7.9 TCRT Infrared

System proposal

Progression in April

1.Correction for last progression

- Switching component in introduction to related technology and descript a few features.
- Add system proposal to the end of the chapter literature review to make and conclusion and more explanation about the next chapter.

2. System analysis and Design

- make a new design pet feeder

CHAPTER 3
ANALYSIS AND DESIGN

3.1 Circuit Design

The flowchart prepared with a vision that the user has to Initiate Blynk application to control and operate Smart Pet Feeder. The application help assign task and set pre-setting to the device for it to automatically operate on its own afterword. Then there will be several option that the user could select and order the device. Firstly, it is two visual button to release pet's food on the plate 100 grams and 200 grams by set the present angle to 0°and turn the servo motor depend on how much food the user wants.

Secondly, visual button that turn on the camera, this button will display the view of the camera through Blynk application.

Thirdly, it is Weight sensor (Load cell) part, the Blynk application will collect input from user to set the weight limit to make the condition later then notified to the application when the food bowl exceed the weight limit.

Fourthly, it is timer part that will allow the user to set time when to release food to the plate. It also assigns the amount of food that will be distributed.

Lastly, is collecting and processing sensors from the device. This part is consisting of two sensors that operate on the device that is Infrared and ultrasonic. Infrared sensor will detect the movement of a pet then activate the water pump and ultrasonic sensor will detect the amount of food that is in the storage.

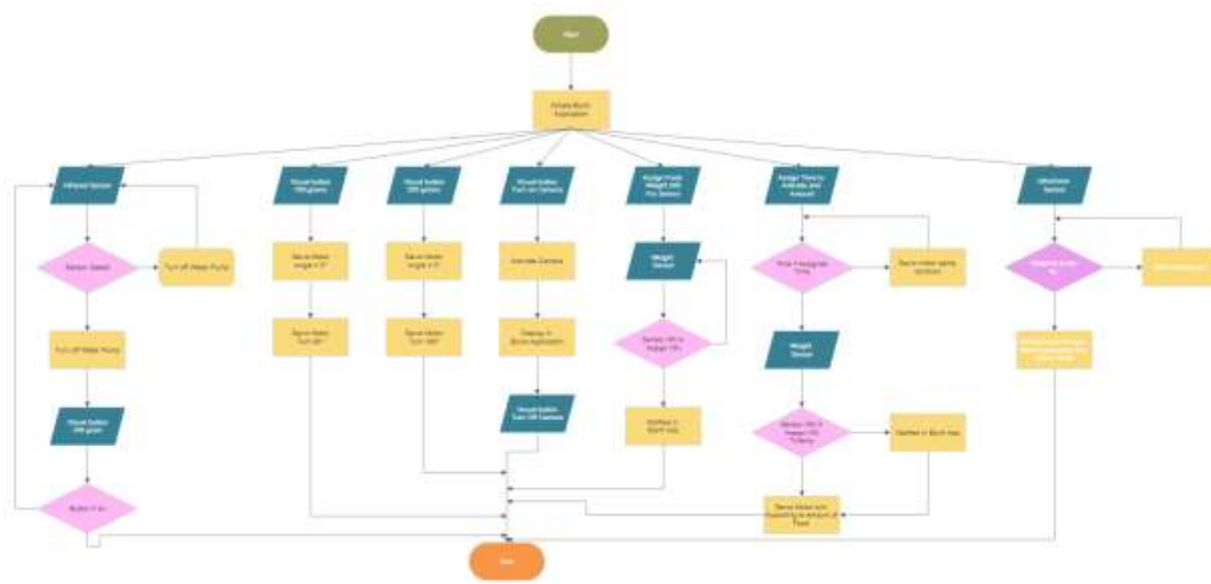


Figure 3.1 System Flowchart

3.2 Electronic Configurations

In addition to fabrication of the storage and feed dish, one of the major components of his project were the electronic configuration and setup. This portion was split into a number of different sections, as follows.

1. Configuration of Load Cell, Amplifier, Power, and esp32 cam.
2. Configuration of Servomotor, Power, and esp32 cam.
3. Configuration of TCRT5000, Power, and esp32 cam.
4. Configuration of water pump, Relay, Power, and esp32 cam.
5. Configuration of ultrasonic sensor, Power, and esp32 cam.
6. Configuration of FT232 USB, Power, and esp32 cam.

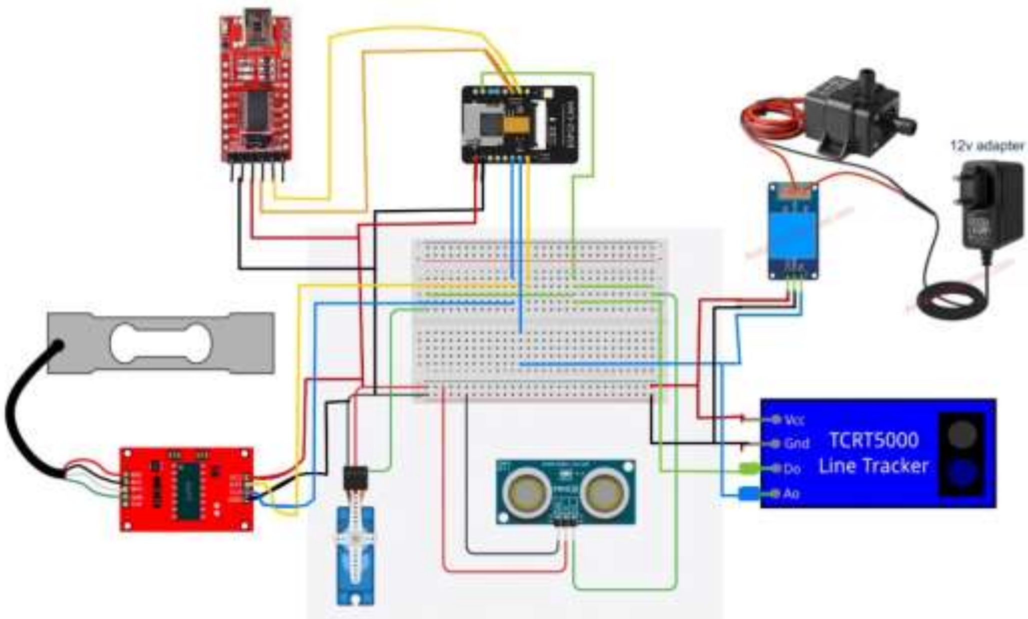


Figure 3.2 Wiring Schematic for the System

3.3 Schematic diagram of auto pet feeder

Automatic pet feeder can contain animal feed 6L/Litters with the size of each side as shown in the Figure 3.3 and according to the picture Figure 3.4 and Figure 3.5 We have plans to install various equipment as follows in Table Equipment on Pet Feeder and Equipment on the Water Fountain.

Figure 3.3.1 Pet Feeder Outer design

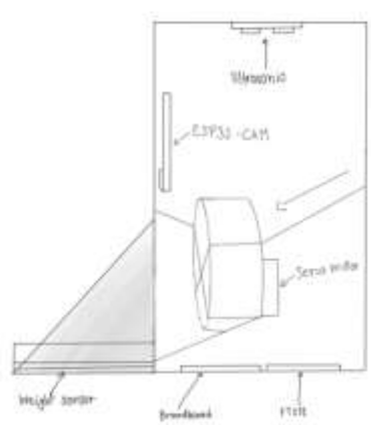


Figure 3.3.2 Pet Feeder inner design

Table 3.1 Equipment On Pet Feeder

Equipment on Pet Feeder		
Material	Position	Purpose
Ultrasonic	paste it on lid of pet feeder.	for sensor amount of animal feed.
ESP32-CAM	paste it above of servo motor following exterior design.	use for viewing pets.
Servo motor	Paste it in the middle of the container.	for delivering food to the food tray.
Weight sensor	Paste it under the tray.	For weighing animal feed.
Breadboard	Paste under the container	for connecting all circuits and materials.
FT232	Paste under the container. Same position as the Breadboard.	For connect and paste code with the computer.

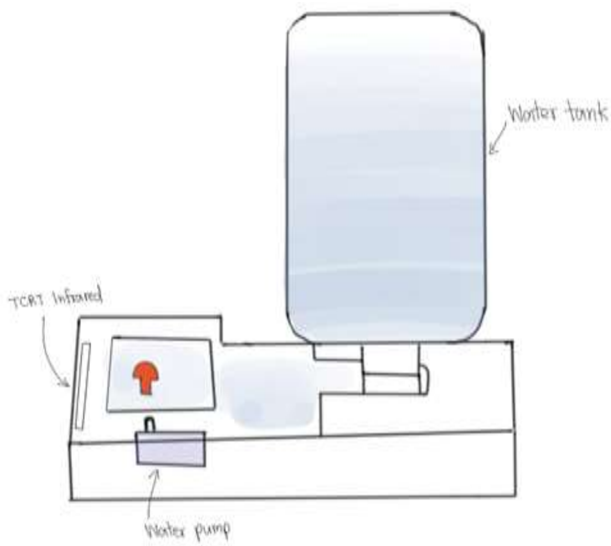


Figure 3.5 Pet Feeder Outer Design Water Fountain

Table 3.2 Equipment On Water Fountain

Equipment on Water Fountain		
Materials	Position	Purpose
Water tank		kept for water storage
TCRT Infrared	In front of cat fountain	Used to detect the distance of pets when they come close to the sensor. Then the cat fountain works.
Water pump	Under cat fountain	to use for making a cat fountain

CHAPTER 4
CONCLUSION

4.1 Conclusion

AUTOMATIC PET FEEDER AND PET MONITORING SYSTEM THROUGH Wi-Fi is a device that will hand out the food to pets. This device has many functions that will help pet's owners feed and look after the pets for example Camera, Ultrasonic sensor, Infrared sensor est. Pet feeder can be controlled, monitored and set up device manually and automatically through Blynk application. This device also allows the pet's owner to feed the food appropriately according to pet's type and weight.

4.2 Discussion

At the present time this project has not tested the device with the pet yet due to incomplete device. However, this project has progress smoothly and waiting for the equipment to arrive and continuing the project. Currently, the project has a slight problem with the Blynk application, the application has so many limited function that we ca use for our device, in order to fix this problem is to purchase the pro version of the application.

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